Software Architecture (WS 26/27)
Assignment 1

Prof. Dr.-Ing. Ronny Seiger (ronny.seiger@rwth-aachen.de)
Tobias Arens (tobias.arens@rwth-aachen.de)

15.10.2026

Task 0: Organize yourselves in groups

One key part of the course is a semester-long group project. In this project you will incrementally build a software system and investigate software architecture-related aspects. Software development is group work: we ask you to organize yourselves into **groups of four** to work on a common project. The group constellations will stay the same during the entire semester.

Task: Sign up for the pre-configured groups in Moodle by Thursday, October 15 (end of day, Aachen time). Use the forum in Moodle to find group members. Sign up to the **AutoFill** group if you want the teachers to assign you to an existing group. We will start adding more students from the waiting list to the course to allow them to join the existing groups afterwards. If you are not part of a group, you will not be able to complete the group project homework and thus, not be admitted to the written exams.

Task 1: Create git Repository

1. Login / Sign-up at the NRW gitlab instance: https://gitlab.git.nrw/
2. (only one in the group) Create a new blank project and name it: **swa-26-<moodle-group-id>-<groupname>**. You can choose your group's name freely.
3. Add all group members to the project:
   - When in your project click in the left sidebar `Manage` -> `Members`
   - At the top right: `Invite members`
   - You can then enter the other's usernames (of the form `@tobias.arens`).
   - Every member should have at least the role Maintainer.
4. Add both `@ronny.seiger` and `@tobias.arens` to your project as Maintainer.
5. We recommend setting up an SSH-key[1] for easier access.

[1] https://docs.gitlab.com/user/ssh/?tab=Git+Bash+on+Windows

Task 2: Brainstorm your project idea

Your are free to choose the concrete domain for your group projects. Every week[2], we will provide new aspects in the exercise sheets that you should add to or investigate in your projects. We will start with simple, standalone software components (services), and add more structured services in the following weeks. One of the services should also interact with an LLM via RWTH KI:connect. Later, these services will have to interact via their APIs in different ways and will have to persist data. Finally, a workflow engine will be used to orchestrate more complex workflows among the existing services.

Task: Find a domain for your project. Think about suitable workflows (business processes) that involve the interactions of different standalone services and potentially also human users.

Two good examples for orientation are the *Leasing Ninja*[3] and the *Flowing Retail*[4] projects. Your domains might for example relate to study/university worfklows, personal interests and hobbies (e.g., local clubs or associations), event management (e.g., book Metallica for Louder than Karl 2027), or professional/organizational workflows. They should be significantly different from the two example projects.

[2] In week 1, you do not have to implement anything, yet!

[3] `https://leasingninja.io/`

[4] `https://github.com/berndruecker/flowing-retail`